

Shuo Tao

PHD · APPLIED PHYSICS

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Education

Rutgers University-Newark

PHD, APPLIED PHYSICS

- Principal Investigator & Research Advisor: Dr. Li Zhu

Newark, New Jersey

09/2020 - 05/2025

SUNY at Stony Brook

MA, PHYSICS

- Research Advisor: Dr. Alexander G. Abanov
- Research Advisor: Dr. Marivi Fernández-Serra

Stony Brook, New York

09/2017 - 05/2020

Tamkang University

BS, PHYSICS

- 2013 Mainland China Students Entrance Scholarship
- Excellent Academic Performance Awards in 2013-2016 academic years

New Taipei City, Taiwan

09/2013 - 06/2017

Professional Experience

2025-2026 **Postdoc Research Fellow**, Mechanical Engineering, The University of North Carolina at Charlotte

2024-2025 **Graduate Fellow**, Graduate School-Newark, Rutgers University-Newark

2023-2024 **Graduate Research Assistant**, Physics Department, Rutgers University-Newark

2020-2023 **Graduate Teaching Assistant**, Physics Department, Rutgers University-Newark

Publications

PUBLISHED

Shuo Tao, and Li Zhu, Machine-Learning-Guided Discovery of Direct-Gap Si-Ge Alloys in Hexagonal and Orthorhombic Silicon Allotropes

Comput. Matter. Today, **9** 100045, 2025

Shuo Tao, Ying Sun and Li Zhu, Computational Discovery of a Novel Yttrium Silicide as an Ambient Pressure Superconductor,

Ann. Phys. (Berl.), e00210, 2025

Shuo Tao, and Li Zhu, EOSnet: Embedded Overlap Structures for Graph Neural Networks in Predicting Material Properties

J. Phys. Chem. Lett., **16** 717, 2025

Julia-Maria Hübner, Thomas Shiell, Piotr Guńka, **Shuo Tao et al.**, A Sodium Germanosilicide with Unusual Network Topology

J. Am. Chem. Soc., **146** 20544, 2024

Shuo Tao, Xuecheng Shao, and Li Zhu, Accelerating Structural Optimization through Fingerprinting Space Integration on the Potential Energy Surface

J. Phys. Chem. Lett., **15** 3185, 2024

Shuo Tao, and Li Zhu, Route to A Direct-gap Silicon Allotrope Si₃₂

J. Phys.: Condens. Matter, **34** 154006, 2022

IN SUBMISSION

Greeshma C. Jose, James L. Petri, **Shuo Tao**, Li Zhu, Weiwei Xie, Sang-Wook Cheong, and Wenli Bi, High-pressure Studies of Superconductivity and Crystal Structure in Mo₃Al₂C, *Phys. Rev. B* (2026)

Shuo Tao, and Qiang Zhu, Accelerated Inorganic Electrides Discovery by Generative Models and Hierarchical Screening, *Phys. Rev. X Intelligence* (2026, under revision) [preprint: arXiv:2601.21077]

Honors & Grants

2024	Dean's Dissertation Fellowship , Rutgers University-Newark	\$ 30,000
	Graduate School-Newark Travel Grant , Rutgers University-Newark	\$ 665
2023	Graduate School-Newark Travel Grant , Rutgers University-Newark	\$ 900
2022	Research Week Three-Minute Poster Presentation Contest, Graduate Student Award, 2nd Place , Rutgers University-Newark	\$ 300
	Graduate School-Newark Travel Grant , Rutgers University-Newark	\$ 660
	2022 Academic Year Teaching Assistant Excellence Award , Physics Department, Rutgers University-Newark	
2013-2016	Excellent Academic Performance Awards , Physics Department, Tamkang University	\$ 4,000
2013	Mainland China Students Entrance Scholarship , Tamkang University	\$ 2,000

Teaching Experience

2023	Algebra Based Physics 21:750:203:01 , Physics Department, Rutgers University-Newark
	Algebra Based Physics 21:750:203:02 , Physics Department, Rutgers University-Newark
	Intro Physics Lab 21:750:206:02 , Physics Department, Rutgers University-Newark
2022	Intro Physics Lab 21:750:205:09 , Physics Department, Rutgers University-Newark
	Intro Physics Lab 21:750:206:11 , Physics Department, Rutgers University-Newark
2021	Intro Physics Lab 21:750:205:04 , Physics Department, Rutgers University-Newark
	Intro Physics Lab 21:750:206:08 , Physics Department, Rutgers University-Newark
2020	Intro Physics Lab 21:750:205:04 , Physics Department, Rutgers University-Newark

Outreach & Development

SERVICE AND OUTREACH

2021-2024	Physics Graduate Student Club , Member, Rutgers University-Newark
2017-2020	Graduate Badminton Club , Secretary, SUNY at Stony Brook

DEVELOPMENT

2024 Gordon Research Conference | Research at High Pressure, Poster Presenter
(FpGNN: A Fingerprint Embedded Graph Neural Network for Material Property Prediction)

2023 American Physical Society March Meeting, Oral Speaker
(M59 - Route leading to exotic silicon allotropes and compounds | Matter under Extreme Conditions I)

2022 Gordon Research Conference | Research at High Pressure, Poster Presenter
(Open-framework allotrope of silicon with higher photon absorption rate)

PROFESSIONAL MEMBERSHIPS

American Physical Society Graduate Student Membership